VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS302

Database Management Systems & Distributed Transactions - Laboratory Manu

Syllabus Focus: CS302 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Relational Formalisms & Schema Design

" Syllabus Scope: Relational algebra operators, tuple relational calculus, ER-to-relational schema mappin

Unit 2: Structured Query Language SQL

" Syllabus Scope: Complex joins, correlated subqueries, aggregate grouping, window functions, views, trig

CS302: Database Management Systems & Distributed Transactions - Advanced Topics &

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Functional Dependencies & Normalization

" Armstrong axioms, canonical cover, attribute closure, 1NF, 2NF, 3NF, BCNF, lossless join, dependency preserv

Unit 4: Transaction ACID & Concurrency Control

" Serializability, conflict graph testing, Two-Phase Locking 2PL, Strict 2PL, timestamp ordering, deadlock det

Unit 5: Storage Engines, Indexing & Query Tuning

" B+ Tree index insertion and deletion, hash indexes, query execution trees, cost-based optimizer estimates

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Enterprise Schema Definition with Strict Foreign Keys and Check Constraints in PostgreSQL

Lab 2: Complex Analytical SQL Queries using Subqueries, Grouping Sets, and Window Functions

Lab 3: Database Event Triggers and Stored Procedures for Automatic Audit Logging

Lab 4: Concurrent Transaction Isolation Testing: Demonstrating Dirty Reads and Lost Updates

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Raghu Ramakrishnan, Johannes Gehrke - Database Management Systems McGraw-Hill 3rd Ed

* Abraham Silberschatz, Henry F. Korth - Database System Concepts McGraw-Hill 7th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students